Exercise Sheet no. 1 · Solutions to exercises 4–7

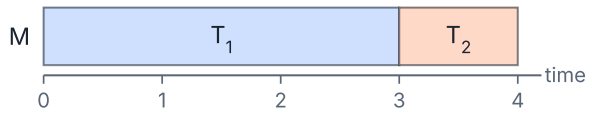
Thor (BigDataA, S1) · USTHB · built from the class solutions and extended with explanations

Exercise 4 · $\overline{L}$ and $\overline{T}$ are not equivalent

One machine, $r_i = 0$. $p = (3, 1)$, $d = (3, 4)$. Compute $\overline{L}$ and $\overline{T}$ for both schedules.

Formulas: $l_i = c_i - d_i$, $t_i = \max\{l_i, 0\}$, $\overline{L} = \Sigma l_i/n$, $\overline{T} = \Sigma t_i/n$.

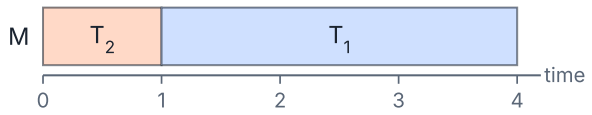
S1 = (T_1, T_2)



	T_1	T_2
c_i	3	4
$l_i = c_i - d_i$	0	0
$t_i = \max\{l_i, 0\}$	0	0

$\overline{L}_1 = (0 + 0)/2 = \mathbf{0}$
$\overline{T}_1 = (0 + 0)/2 = \mathbf{0}$

S2 = (T_2, T_1)



	T_1	T_2
c_i	4	1
$l_i = c_i - d_i$	1	-3
$t_i = \max\{l_i, 0\}$	1	0

$\overline{L}_2 = (1 - 3)/2 = \mathbf{-1}$
$\overline{T}_2 = (1 + 0)/2 = \mathbf{0.5}$

	S1	S2	best
$\overline{L}$	0	-1	S2
$\overline{T}$	0	0.5	S1

S2 is optimal for $\overline{L}$ ($-1 < 0$) but not for $\overline{T}$ ($0.5 > 0$). S1 is optimal for $\overline{T}$ but not for $\overline{L}$. So a schedule optimal for one is not necessarily optimal for the other: the criteria are not equivalent.

Why: lateness lets an early task ($l_2 = -3$) cancel a late one in the sum; tardiness clips early tasks at 0, so it cannot.

Exercise 5 · Optimal for L_{max} ⇒ optimal for T_{max}

Show that a schedule optimal for L_{max} is also optimal for T_{max}.

$L_{max} = \max l_i = \max (c_i - d_i)$ and $t_i = \max\{l_i, 0\}$. Hence

$$T_{max} = \max t_i = \max_i \max\{l_i, 0\} = \max\{ \max_i l_i, 0 \} = \mathbf{\max\{L_{max}, 0\}}$$

So $T_{max} = g(L_{max})$ with $g(x) = \max\{x, 0\}$, and g is non-decreasing. Let S^* be optimal for L_{max}: $L_{max}(S^*) \le L_{max}(S)$ for every schedule S. Applying g:

$$T_{max}(S^*) = \max\{L_{max}(S^*), 0\} \le \max\{L_{max}(S), 0\} = T_{max}(S)$$

So S^* is optimal for T_{max}. ∎

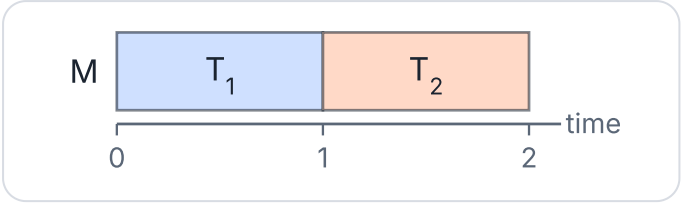
- If $T_{max} > 0$, we can deduce L_{max} ($= T_{max}$).
- If $T_{max} = 0$, we cannot: we only know $L_{max} \leq 0$.
- From L_{max} we can always deduce T_{max}, but not the other way round. So the converse is not necessarily true (Exercise 6).

Exercise 6 · Optimal for T_{max} but not for L_{max}

Give an example (two tasks, one machine) where an optimal solution for T_{max} is not optimal for L_{max}.

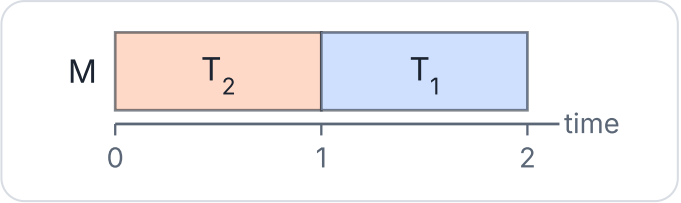
T_i	T_1	T_2
p_i	1	1
d_i	5	10

(T_1, T_2)



$c = 1, 2 \cdot l = -4, -8$
$L_{max} = \mathbf{-4} \cdot T_{max} = \mathbf{0}$

(T_2, T_1)



$c_2 = 1, c_1 = 2 \cdot l_2 = -9, l_1 = -3$
$L_{max} = \mathbf{-3} \cdot T_{max} = \mathbf{0}$

Both schedules have $T_{max} = 0$, so (T_2, T_1) is optimal for T_{max}. But its $L_{max} = -3$ is worse than the -4 of (T_1, T_2), so it is not optimal for L_{max}.

Why: when every task is on time, $T_{max} = \max\{L_{max}, 0\} = 0$ whatever the (negative) L_{max}, so T_{max} cannot tell these schedules apart.

Exercise 7 · Classify the maintenance workshop

5 platforms, one vehicle each. Inspection duration depends on the vehicle and the type of inspection. Minimise the average waiting time of the vehicles.

P5 | r_i | $\overline{F}$

- **α = P5.** 5 platforms, each able to hold one vehicle, and any vehicle fits any platform. The duration depends on the vehicle and inspection only, not on the platform: $p_{ij} = p_i$. So the machines are **identical parallel** (P), exactly 5. Not R: R is for durations that also depend on the machine.
- **β = r_i.** Each vehicle comes to the workshop after its own period in service, so vehicles become available at different dates: release dates. No preemption, no precedence, no due dates, durations arbitrary: nothing else to write.
- **γ = $\overline{F}$.** The waiting time of a vehicle is its flow time minus its inspection time, $f_i - p_i$. The mean of the p_i is a constant, so minimising the average waiting time is the same as minimising the mean flow time $\overline{F}$.

Remark: $\overline{F} = \overline{C} - (\Sigma r_i)/n$ (Exercise 3), so P5$|r_i|\overline{C}$ is an equivalent way to write it.